

Finding Periods

Time limit: 1.00 s **Memory limit:** 512 MB

A *period* of a string is a prefix that can be used to generate the whole string by repeating the prefix. The last repetition may be partial. For example, the periods of `abcbabca` are `abc`, `abcbabc` and `abcbabca`.

Your task is to find all period lengths of a string.

Input

The only input line has a string of length n consisting of characters `a–z`.

Output

Print all period lengths in increasing order.

Constraints

- $1 \leq n \leq 10^6$

Example

Input:
`abcbabca`

Output:
`3 6 7`